

⇒ Johnson - Trotter

```
#include <stdio.h>
```

```
void printPerm(int perm[], int n) {
    for (int i = 0; i < n; i++)
        printf("%d ", perm[i]);
    printf("\n");
}
```

```
int getMobile(int perm[], int dir[], int n) {
    int mobile = 0;
    int mobile_idx = -1;
    for (int i = 0; i < n; i++) {
        if (dir[perm[i]] == -1 || perm[i] > mobile) {
        mobile = perm[i];
        mobile_idx = i;
    }
}
```

```
if (dir[perm[i]-1] == -1 || i == 0) {
    if (perm[i] > perm[i-1] || perm[i] > mobile) {
        mobile = perm[i];
        mobile_idx = i;
    }
}
```

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```
if (dir[perm[i]-1] == 1 || i == n-1) {  
    if (perm[i] > perm[i+1] && perm[i] > mobile) {  
        mobile = perm[i];  
        mobile_idx = i;  
    }  
}  
}  
return mobile_idx;  
}
```

```
void JohnsonTrotter (int n) {
```

```
    int perm[n];
```

```
    int dir[n];
```

```
    for (int i = 0; i < n; i++) {
```

```
        perm[i] = i + 1;
```

```
        dir[i] = -1;
```

```
    }
```

```
    printPerm(perm, n);
```

```
    while (1) {
```

```
        int mobile_idx = getMobile(perm, dir, n);
```

```
        if (mobile_idx == -1) break;
```

```
        int mobile = perm[mobile_idx];
```

```
if (dir[mobile - 1] == -1) {
    int temp = perm[mobile_idx];
    perm[mobile_idx] = perm[mobile_idx + 1];
    perm[mobile_idx + 1] = temp;
    mobile_idx++;
}
```

```
} else {
    int temp = perm[mobile_idx];
    perm[mobile_idx] = perm[mobile_idx + 1];
    perm[mobile_idx + 1] = temp;
    mobile_idx++;
}
```

```
for (int i = 0; i < n; i++) {
    if (perm[i] > mobile)
        dir[perm[i] - 1] += -1;
}
```

```
printPerm(perm, n);
}
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    printf("Enter no. of elements: ");
```

```
    scanf("%d", &n);
```

johnsonTrotter(n);

return 0;

}

o/p: Enter no. of elements : 3

1 2 3

1 3 2

3 1 2

3 2 1

2 3 1

2 1 3

Beit
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